Cheatsheet AZ-305 — 11 · Costs, purchasing models and composite SLAs

1. Compute savings models

Mechanism	What it is	Savings	Commitment	Applies to	When
Pay-as-you-go	Per second/hour with no commitment	0 %	None	Everything	Unpredictable or temporary workloads
Reserved Instances (RI)	1- or 3-year reservation of **one VM family in one region**	Up to ~72 %	High: fixed family + region (size flexibility within the family)	VMs, SQL DB/MI vCore, Cosmos DB, Storage, App Service, Synapse, Redis…	**Stable and predictable** workload on specific compute
Savings Plan for compute	**Hourly spend** commitment (€/h) for 1 or 3 years	Up to ~65 %	Medium: **any region, family or compute service**	VMs, App Service, Functions Premium, Container Apps, Container Instances, Dedicated Hosts	Stable workload that **changes in type or region**
Spot VMs	Spare capacity, **evictable** with 30 s notice	Up to ~90 %	None	VMs, VMSS, AKS node pools	**Interruptible** workloads: batch, render, test, CI; never critical production
Azure Hybrid Benefit	Use **Windows Server / SQL Server licenses with Software Assurance** (or subscription)	Up to ~40 % Windows, ~55 % SQL (85 % combined with RI)	Own licenses	Windows VMs, SQL on VM, Azure SQL DB/MI **vCore**, AKS Windows nodes, Azure Stack HCI	"Software Assurance", "License Mobility", "existing licenses"
Dev/Test pricing	Reduced rates with no Windows license in Dev/Test subscriptions (Visual Studio)	Variable	—	Non-production environments	"development and test environment"
Autoscale / Serverless / scale to zero	Pay only for what is used	Variable	—	VMSS, App Service, Functions, Container Apps, SQL Serverless	Variable workload with troughs
Scheduled shutdown	Automation runbook / VM Auto-shutdown	100 % of the time switched off	—	Dev/test VMs	Business hours

- RI and Savings Plan **combine** (RI is applied first). Both are **billing** constructs, they do not reserve physical capacity (for that: **capacity reservations** or Dedicated Hosts).
- **Reserved capacity** does not exist with **Serverless** (SQL) or with Spot.
- "Minimize costs" in an exam question means **the cheapest option that meets the technical requirements**, never the cheapest in absolute terms.

2. Storage and data savings

Lever	Effect
Access tiers Hot → Cool → Cold → Archive + lifecycle management	Lower cost per GB, higher access cost/latency
LRS < ZRS < GRS < RA-GRS < GZRS < RA-GZRS	Fewer copies = lower cost; pick the minimum that protects what is required
Standard vs Premium	Premium only if latency/IOPS are required
Storage reserved capacity (100 TB / 1 PB, 1-3 years)	Discount on Blob/ADLS
SQL: elastic pool (share), Serverless (pause), reservations, Hybrid Benefit, Standard vs Premium	See cheatsheet 09
Log Analytics: commitment tiers, Basic/Auxiliary Logs, retention, daily cap	See cheatsheet 07
Managed disks: Standard HDD < Standard SSD < Premium SSD < Premium v2 < Ultra	Pick by required IOPS/latency

3. Cost Management + Billing

Tool	What it does	Signal
Cost analysis	Explore and group costs (by tag, RG, service, subscription)	"review weekly costs", "group by department" → **tags**
Budgets	Spending threshold with alerts (email / action group); does not block spending	"notify when spend exceeds"
Cost alerts	Budget, anomalies, credit	"unexpected increase" → anomaly alerts
Advisor (Cost)	Underutilized VMs, orphaned disks, recommended RIs; **does not act**	"recommendations"
Pricing calculator / TCO calculator	Estimate the cost of an architecture / compare with on-prem	"estimate before migrating"
Exports	Scheduled dump of costs to Storage	"analyze in Power BI"
Management groups + subscriptions	Billing structure (EA: Enrollment → Departments → Accounts → Subscriptions; MCA: Billing profiles → Invoice sections)	"chargeback"
Mandatory tags (Policy)	Guarantee cost dimensions	"ensure every resource has a cost center"
Cost Management Reader / Contributor	RBAC to view / manage budgets	"auditors must review costs" → Reader

4. Service SLAs (the ones you must know)

Service	SLA
Single VM with Premium SSD / Ultra	99.9 %
Single VM with Standard SSD	99.5 %
Single VM with Standard HDD	95 %
VMs in an **Availability Set**	99.95 %
VMs in **Availability Zones** (≥2)	99.99 %
App Service (Basic+)	99.95 %
Azure SQL Database / MI	99.99 % (99.995 % Business Critical zone-redundant)
Cosmos DB	99.99 % single-region; 99.999 % multi-region reads
Storage (LRS/ZRS/GRS)	99.9 % read/write (RA-GRS read 99.99 %)
Load Balancer Standard	99.99 %
Application Gateway v2	99.95 %
Front Door	99.99 %
Traffic Manager	99.99 %
VPN Gateway	99.9 % (Basic) / 99.95 % (Standard+)
ExpressRoute	99.95 %
AKS (control plane, with uptime SLA)	99.95 % (99.9 % without zones)
Key Vault	99.99 %
Functions Premium / App Service	99.95 %

5. Composite SLA calculation

In series (all must work): multiply.

```
Web (99.95 %) → SQL (99.99 %)  =  0.9995 × 0.9999 = 0.9994 → 99.94 %
```

Each component in series **lowers** the total SLA. One component at 99.9 % drags the whole thing below 99.9 %. **In parallel (one is enough):** 1 − product of the unavailabilities.

```
Two independent regions at 99.95 %:  1 − (0.0005 × 0.0005) = 0.99999975 → 99.99997 %
```

Adding redundancy **raises** the SLA above the individual component. **Mixes:** first compute the parallel blocks, then multiply the series.

```
Traffic Manager (99.99 %) → [Region A ‖ Region B], each region = App (99.95 %) × SQL (99.99 %) = 99.94 %
Parallel: 1 − (0.0006)² = 99.99996 %
Total: 0.9999 × 0.9999996 ≈ 99.99 %
```

Exam rules:

- "SLA 99.99 % per tier" with VMs → **Availability Zones** (Availability Set only gives 99.95 %).
- Adding a load balancer or gateway in series **lowers** the total; it must be compensated with parallelism.
- Minutes of unavailability per month: 99.9 % ≈ 43 min · 99.95 % ≈ 22 min · 99.99 % ≈ 4.3 min · 99.999 % ≈ 26 s.
- The SLA of the weakest component in series is the ceiling of the total.

6. Decision tree (compute savings)

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What is the workload like?
├─ Stable, fixed compute, same region ..... Reserved Instances (+ Hybrid Benefit if there are licenses)
├─ Stable but changes family/region ..... Savings Plan
├─ Interruptible (batch, test, render) ..... Spot
├─ Variable with troughs ..... Autoscale / Serverless / scale to zero
├─ Dev/test during business hours ..... Auto-shutdown + Dev/Test pricing
└─ I have licenses with Software Assurance ..... Hybrid Benefit (vCore in SQL)
```

7. Signals → answer

Wording	Answer
"predictable workload, 3 years, same VM size"	Reserved Instances
"commit to spend, flexibility across regions and VM families"	Savings Plan
"batch job can be interrupted, lowest cost"	Spot VMs
"Software Assurance", "existing licenses"	Azure Hybrid Benefit
"auditors review costs weekly"	Cost Management (Cost Management Reader)
"alert when spending exceeds"	Budget

Wording	Answer
"group costs by department"	Tags + Cost analysis
"SLA 99.99 % for VMs"	Availability Zones
"estimate cost of migration vs on-prem"	TCO calculator
"Serverless" + "reserved capacity"	Incompatible → elastic pool

8. Mistakes to avoid

1. Spot for production or non-interruptible workloads.
2. RI when the workload changes family/region → Savings Plan.
3. Hybrid Benefit in the DTU model → vCore only.
4. Availability Set when 99.99 % is required.
5. Adding SLAs instead of multiplying them in series.
6. Choosing the cheapest option that does not meet the technical requirement.